MEGAN DAOUST

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EDUCATION

University of Quebec in Outaouais **January 2023 - December 2026**
Bachelor's, Computer Science *GPA: 9.3*

- 2026 Excellence Scolarship Recipient
- Participated in Quebec Hackfest 2024 and 2025
- Student Representative of the Computer Science Department Council

SKILLS

Languages: JavaScript, TypeScript, iOS/Swift, Python, Java, Bash, C/C++, SQL
Cloud & Infrastructure: AWS (EC2, RDS), Nginx, Docker, Github Actions, Azure DevOps
Web Technologies: React.js, Node.js, Django, Flask, HTML/CSS, SwiftUI
Databases: PostgreSQL, Supabase
Networking & VoIP: TCP/IP, OSI Model, Firewalls (IP Tables), Subnetting, IP Routing (Static & Dynamic), Routing Protocols (OSPF, RIP), VoIP, Asterisk
Other: Git, Gitlab, JIRA, Postman, Cisco Packet Tracer, Wireshark

PROFESSIONAL EXPERIENCE

Public Safety Canada **May 2026 - Present**
Software Developper Intern *Ottawa, ON, Canada*

- Developping Chrome web extensions leveraging various APIs to facilitate multidisciplinary teams work more efficiently day-to-day.
- Engineered a secure, full-stack internal web application from architectural design to deployment, utilizing a modern frontend library for a responsive user interface and a scalable backend API.
- Architected and managed production-grade cloud infrastructure, configuring cloud compute instances for application hosting, a managed relational database service for data persistence, and a high-performance web server as a reverse proxy.
- Integrated specialized telephony solutions by implementing VoIP concepts and configuring open-source PBX communication platforms to modernize internal communication workflows.

Public Service Commission of Canada **January 2026 - May 2026**
Software Developer Intern *Ottawa, ON, Canada*

Software Developer Intern **April 2025 - August 2025**

- Contributed to the development and maintenance of web applications using React (frontend) and Django (backend), while applying best coding practices.
- Performed advanced SQL queries using Microsoft SQL Server Management Studio to retrieve and manipulate relational database data.

Employment and Social Development Canada **September 2025 - December 2025**
Software Developer Intern *Ottawa, ON, Canada*

Full-Time Data Analyst Intern **January 2025 - April 2025**

- Automate data processing workflows using Python and scripting tools to improve efficiency and reduce manual effort.

University of Quebec in Outaouais **September 2025 - December 2025**

Teaching Assistant *Gatineau, QC, Canada*

- Class: Intro to Cybersecurity

PROJECTS & OUTSIDE EXPERIENCE

Cabinet Équilibre - *Link to project* **August 2026 - October 2026**

- Full-Stack Healthcare Booking Web Application (React, Node.js, PostgreSQL, AWS, Stripe)
- Built and deployed an end-to-end, bilingual clinical management and appointment booking platform serving practitioners and clients in production.
- Fortified backend security using AWS RDS IAM dynamic database authentication (@aws-sdk/rds-signer), replacing hardcoded credentials with ephemeral, auto-refreshing SSL tokens.
- Integrated AWS Cognito (OAuth 2.0 / JWT) to establish Role-Based Access Control (RBAC), securing admin scheduling routes and therapist dashboards.

VestiAIre **September 2026 - Present**

- Developed VestiAIre, an iOS wardrobe-management application using Swift, SwiftUI, MVVM, and Swift Concurrency, delivering a polished, fully localized user experience.
- Integrated generative AI to analyze users' available clothing and local weather conditions, generating personalized, context-aware outfit recommendations based on style, color, and garment availability.
- Built a secure Supabase/PostgreSQL backend with authentication, asynchronous CRUD operations, database validation, and Row-Level Security to protect user-specific wardrobe data.

Embedded Systems Project: CO2 Air Quality Monitoring System **November 2025 - December 2025**

- Designed and implemented an embedded CO2 monitoring system from concept to validation, utilizing the FRDM-KL28Z development board (ARM Cortex-M0+) and the SEN0159 electrochemical sensor to measure indoor air quality in real-time.
- Developed a robust C-based software driver to manage the 12-bit ADC configuration, featuring automated hardware-level calibration and a 32-sample averaging algorithm to effectively mitigate signal noise and ensure reliable data acquisition.
- Architected a modular system structured in three distinct layers (Application, Driver, and Hardware), encapsulating complex hardware interactions to expose simplified functional interfaces for efficient application-level data processing.

Better Reads **October 2025 - December 2025**

- Architecture & Technology: Designed a modified MERN-stack-inspired architecture (React 19, TypeScript, Vite 7, Django 4.2, PostgreSQL 17); implemented containerized deployment via Docker/Docker-Compose.
- Project Management: Applied Scrum methodology to deliver an MVP across three Sprints; facilitated ceremonies (Sprint Planning, Daily Scrum) and managed the product backlog.
- Quality & Risk Management: Identified system vulnerabilities (JWT authentication, lack of client-side caching) and proposed quality improvement solutions (React Query, PostgreSQL indexing) during the final project review.